

ISHINOMAKI, Japan

WMO: 475920

Lat: 38.4275N

Lon: 141.2994E

Elev: 141

StdP: 14.62

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.3	25.2	14.2	11.4	29.5	15.9	12.4	30.3	28.7	34.3	26.4	35.7	9.0	330	0.505

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.5	85.2	74.6	82.6	74.2	80.5	73.6	76.9	81.9	75.9	80.7	74.9	79.4	10.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.4	133.7	79.6	74.3	129.0	79.0	73.3	124.4	78.3	40.5	82.1	39.5	80.9	38.6	79.5	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	26.1	22.8	20.1	DB	19.7	91.2	2.5	2.9	17.9	93.2	16.5	94.9	15.1	96.5	13.3	98.6
(5)				WB	18.4	78.3	2.2	1.3	16.8	79.2	15.5	80.0	14.2	80.7	12.6	81.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.8	34.0	34.9	40.3	49.4	58.6	65.2	72.3	75.3	69.4	59.0	47.8	38.5	
	DBStd	15.14	4.26	4.62	5.06	5.67	5.00	4.45	4.99	4.40	5.22	5.34	5.78	5.08	
	HDD50	1774	496	423	302	80	3	0	0	0	0	3	108	359	
	HDD65	4848	961	843	764	469	204	51	4	0	17	197	515	823	
	CDD50	3177	0	0	3	60	271	456	693	786	582	282	43	1	
	CDD65	776	0	0	0	0	7	57	232	321	148	11	0	0	
	CDH74	3596	0	0	0	1	21	104	1023	1878	555	14	0	0	
	CDH80	593	0	0	0	0	3	10	179	344	57	0	0	0	
Wind	WSAvg	9.3	10.3	10.6	10.9	10.5	9.6	8.3	7.6	7.8	8.3	8.9	9.3	10.0	
Precipitation	PrecAvg	44.1	1.5	1.4	2.7	3.5	3.9	4.8	5.9	4.6	6.0	5.0	2.8	1.8	
	PrecMax	57.6	5.2	3.3	6.3	6.7	8.0	10.5	12.6	18.4	17.6	15.2	6.2	7.1	
	PrecMin	32.5	0.2	0.2	0.2	0.3	1.3	2.2	2.0	0.6	0.5	1.0	0.6	0.4	
	PrecStd	6.7	1.3	0.8	1.5	1.8	1.7	1.8	3.0	4.1	3.7	3.4	1.7	1.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.6	52.7	60.1	69.4	76.8	81.0	88.7	90.3	84.9	75.7	64.8	57.5	
		MCWB	43.6	45.5	49.0	54.5	62.9	67.7	75.8	76.2	73.3	65.5	56.8	52.3	
	2%	DB	46.1	48.4	54.9	64.4	71.7	76.8	84.3	86.3	81.4	72.2	62.4	52.9	
		MCWB	40.6	42.5	47.2	53.8	59.4	67.0	74.4	75.4	72.9	64.9	55.9	47.4	
	5%	DB	43.6	45.7	52.0	61.0	68.5	74.0	81.5	83.6	79.1	69.8	60.4	49.8	
		MCWB	38.3	40.0	45.3	52.1	58.7	66.0	73.8	75.0	72.5	63.1	54.7	44.5	
	10%	DB	41.2	43.1	49.5	58.3	66.1	71.7	79.2	81.4	77.2	67.8	58.3	47.1	
		MCWB	36.3	37.8	43.8	51.1	58.3	65.3	73.1	74.6	71.6	61.5	52.8	42.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.9	47.5	51.9	58.9	65.9	71.5	77.6	78.8	76.9	69.9	60.1	53.5
			MCDB	48.8	51.1	56.1	63.8	71.5	76.5	84.1	84.1	80.8	72.7	62.5	56.5
2%		WB	41.3	43.6	49.2	56.2	63.1	69.3	76.2	77.3	75.3	66.9	57.9	48.3	
		MCDB	45.2	47.6	53.2	61.1	67.2	74.0	81.6	82.4	79.2	70.3	61.0	52.0	
5%		WB	38.8	40.7	46.8	54.4	61.5	67.9	75.0	76.4	74.0	64.6	55.7	45.2	
		MCDB	42.8	45.1	50.7	58.7	65.8	71.7	79.9	81.3	78.1	68.3	59.7	49.5	
10%		WB	36.7	38.2	44.5	52.5	59.9	66.4	73.8	75.6	72.4	62.4	53.2	42.3	
		MCDB	40.7	42.5	49.1	56.8	64.3	70.3	78.1	80.3	76.5	66.7	57.4	46.6	
Mean Daily Temperature Range	5% DB	MDBR	11.5	12.4	13.7	14.1	12.3	10.0	9.2	9.5	10.9	13.6	14.2	12.0	
		MCDBR	14.5	15.2	17.1	17.8	16.4	13.5	13.0	12.8	12.0	13.8	16.5	15.5	
		MCWBR	10.9	11.2	12.2	10.2	8.4	6.6	5.8	5.6	6.2	8.8	12.0	12.0	
	5% WB	MCDBR	14.5	14.8	16.1	15.0	13.0	10.7	11.0	10.5	9.7	11.6	15.5	15.6	
		MCWBR	11.6	12.1	13.0	10.3	8.1	6.0	5.4	5.1	5.7	9.1	12.9	13.0	
Clear-Sky Solar Irradiance	taub	0.365	0.409	0.483	0.531	0.552	0.545	0.554	0.540	0.475	0.435	0.398	0.373		
	taud	2.192	2.067	1.911	1.852	1.865	1.961	2.000	2.030	2.178	2.187	2.200	2.197		
	Ebn at Noon	248	251	244	241	239	240	236	236	245	241	234	235		
	Edh at Noon	34	44	56	63	64	58	56	53	43	38	33	31		
All-Sky Solar Radiation	RadAvg	668	908	1198	1488	1599	1544	1408	1392	1156	948	706	571		
	RadStd	37	62	70	142	155	95	213	180	108	97	44	42		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	+1.12	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.68	N/A	N/A	N/A	N/A	

Nomenclature: See separate page